VeerDrishti

VeerDrishti is an AI-powered combat vision system engineered to deliver real-time situational awareness and automated visual threat assessments for ground forces in high-stress tactical operations.

Project Overview

- **Primary Objective**: Automate visual monitoring to reduce cognitive fatigue, prevent fatal human errors, and minimize casualties among military personnel.
- **Deployment Hardware**: Designed for edge deployment on low-cost wearable hardware, such as helmet-mounted smartphones or Raspberry Pi units, allowing reliable performance in signal-denied zones.

Technical Methodology

- **Lightweight Detection Models**: Employs edge-optimized deep-learning architectures like YOLO to ensure low-latency, real-time object classification.
- **Multimodal Spectrum Fusion**: Can merge visible RGB texture data with thermal infrared (IR) edge signals through region-attention mechanisms to maintain detection accuracy in low-light, night, or obscured settings.
- **Generative Data Augmentation**: Utilizes Generative Adversarial Networks (GANs) to synthesize military-specific datasets, addressing data scarcity in tactical vision models.
- **Zero-Shot Threat Identification**: Leverages vision-language models to identify novel or unseen threats using natural language supervision.
- **Augmented Reality (AR) Interface**: Applies an instant AR visual overlay to highlight enemy combatants, weapons, and vehicles on-screen while logging local mission metadata for post-operation auditing.

Feasibility & Global Alignment

- **Triple-Dimension Feasibility**: Assesses technical feasibility via mobile GPU performance, operational feasibility through low cognitive overhead, and economic feasibility by leveraging commercial mobile hardware and open-source frameworks.
- **UN SDG Alignment**: Supports UN Sustainable Development Goal 16 (Peace, Justice, and Strong Institutions) by mitigating operational risk (Target 16.1), retaining mission video logs for legal and operational reviews (Target 16.3), and strengthening national defense capabilities (Target 16.A).